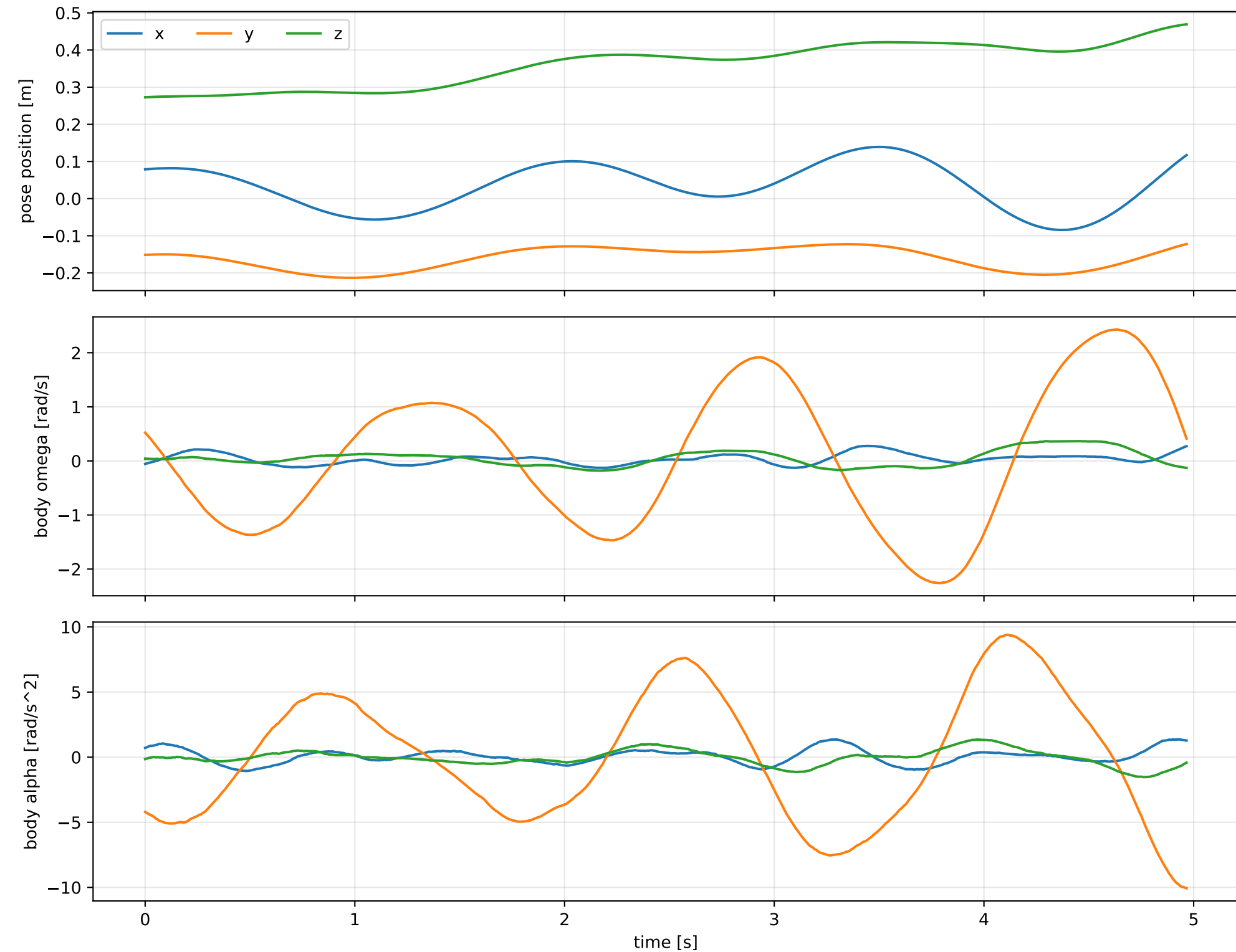
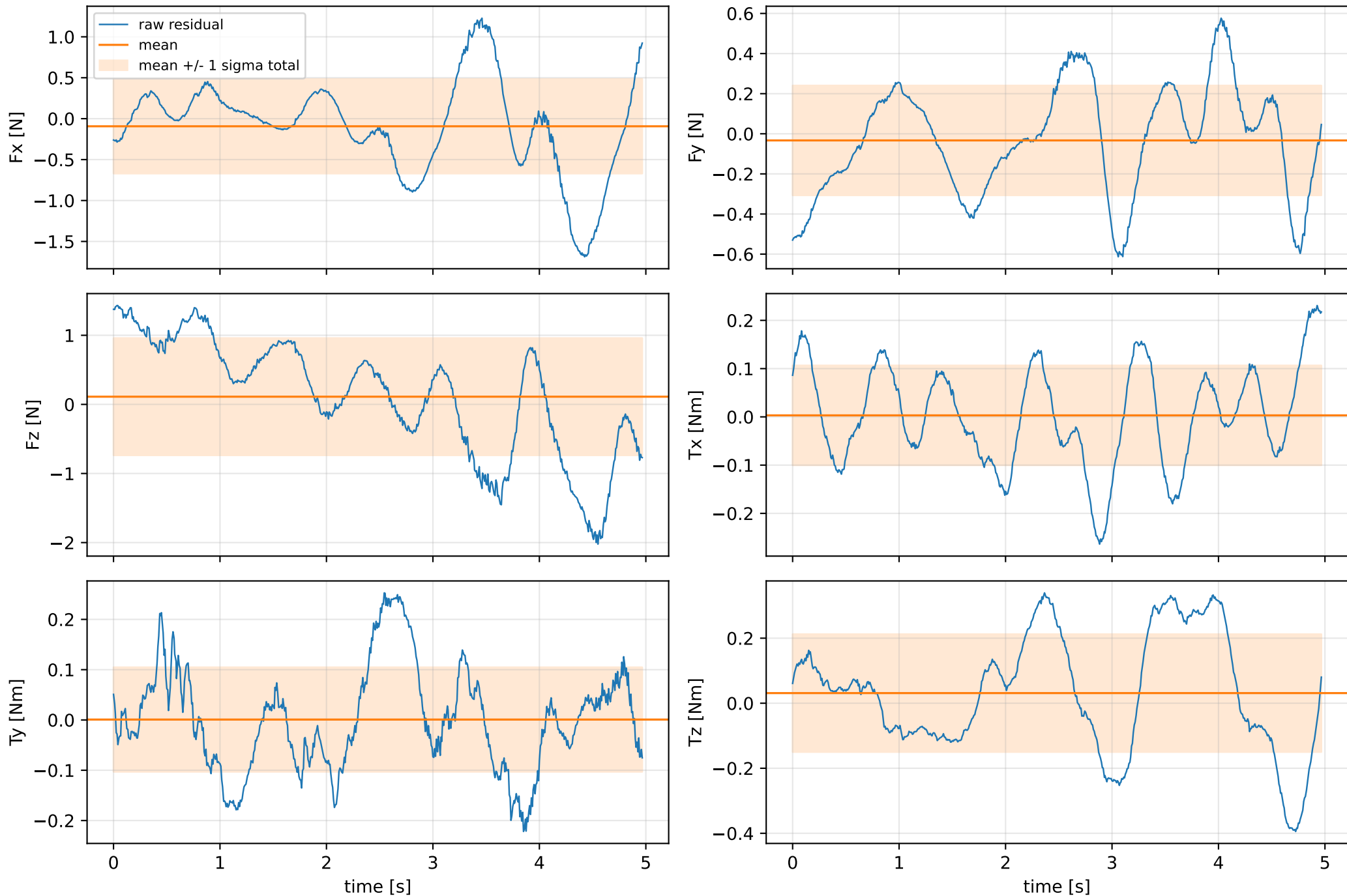


force_over_mass_rotor_3_nominal: SG trajectory and derived motion



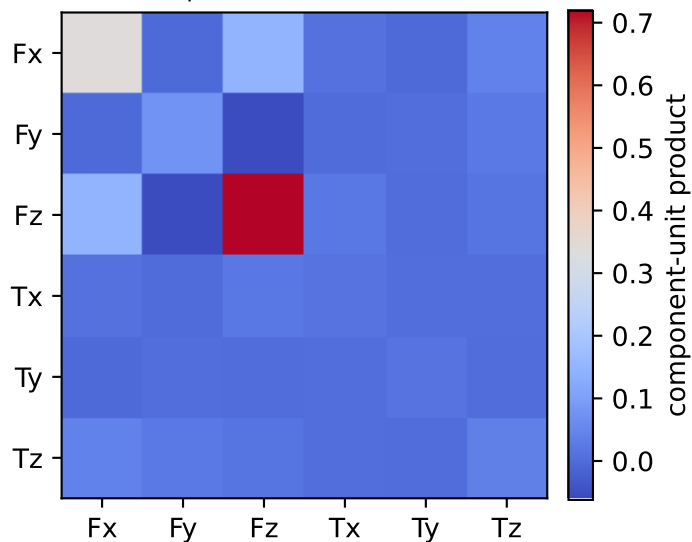
force_over_mass_rotor_3_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



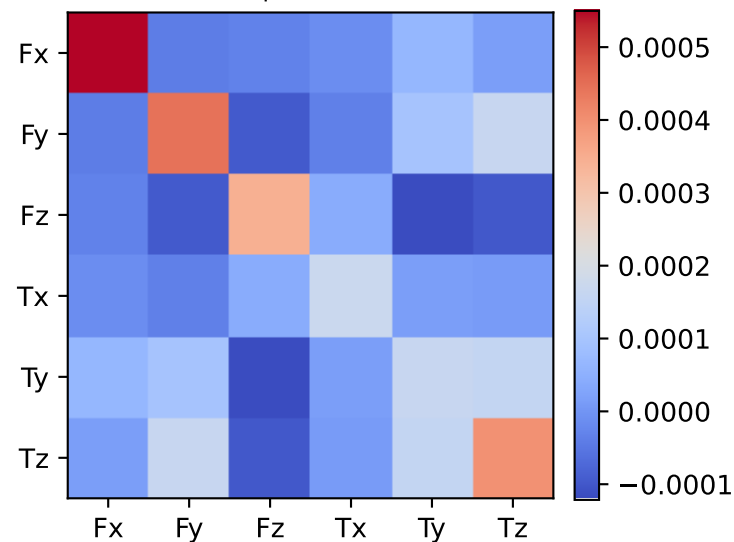
vector RMS about zero: force 1.0711 N, torque 0.23491 Nm; component std about mean: force [0.5773 0.2736 0.8478], torque [0.1034 0.1039 0.1811]

force_over_mass_rotor_3_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.01 0.01 0.018 0.078 0.288 0.775]

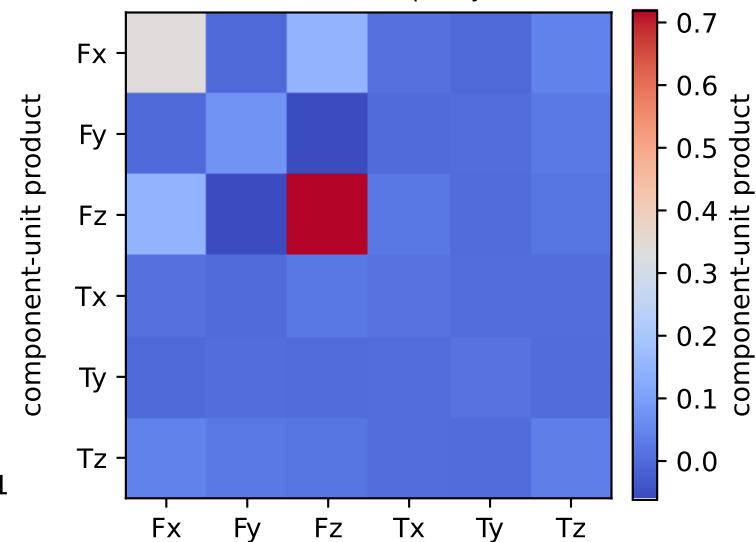
empirical total covariance



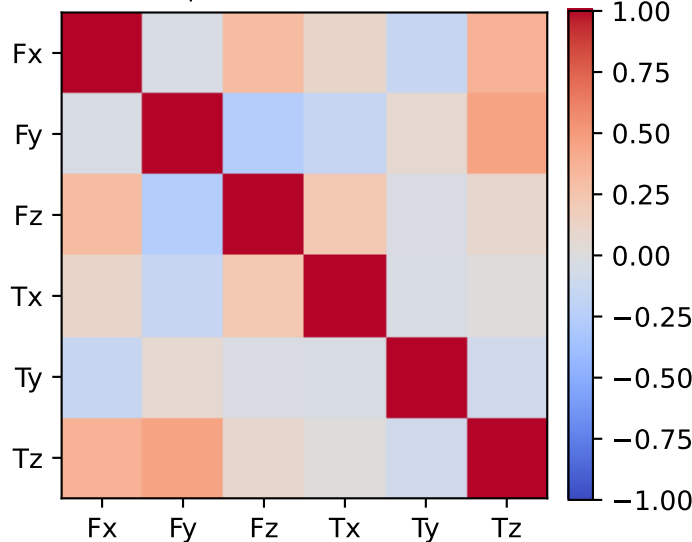
mean full-SG prediction covariance



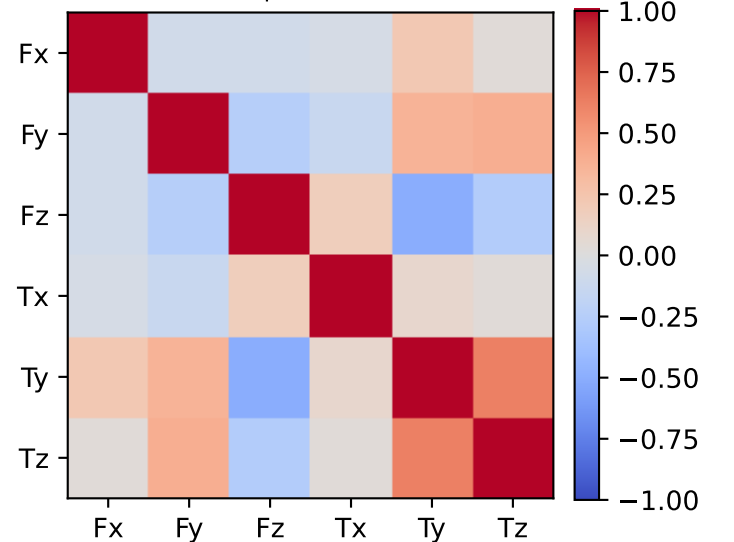
PSD excess model discrepancy covariance



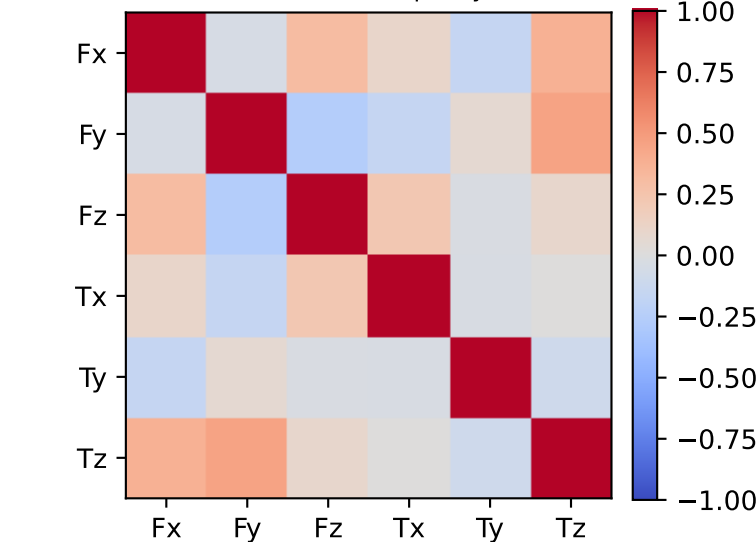
empirical total correlation



mean full-SG prediction correlation

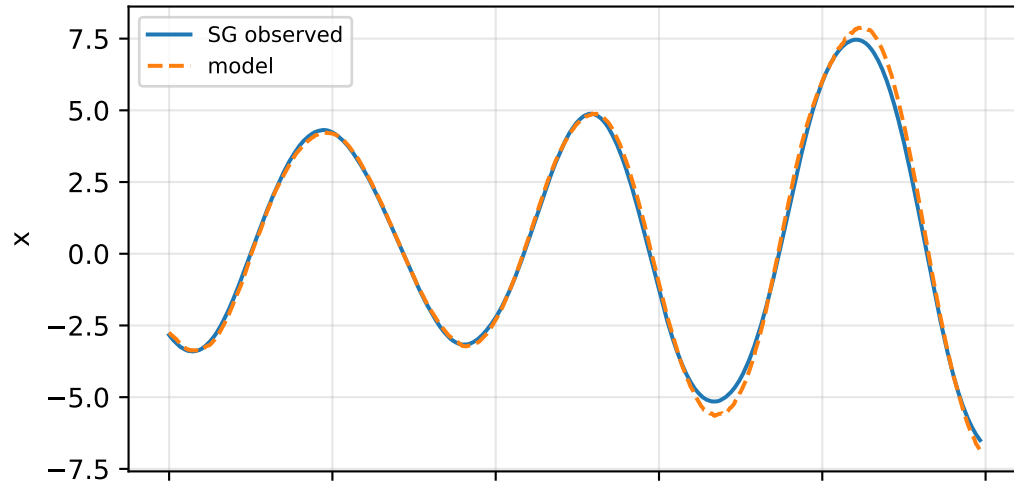


PSD excess model discrepancy correlation

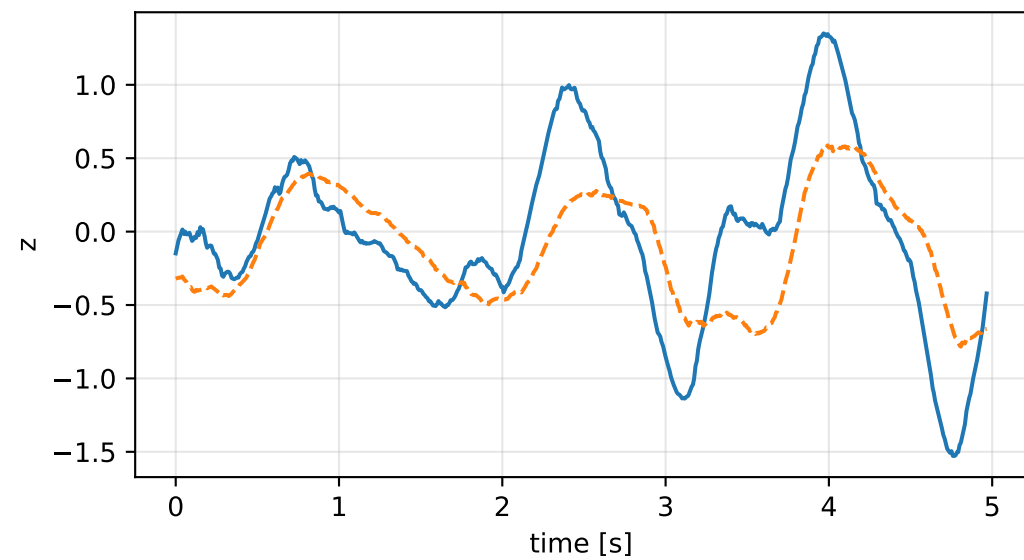
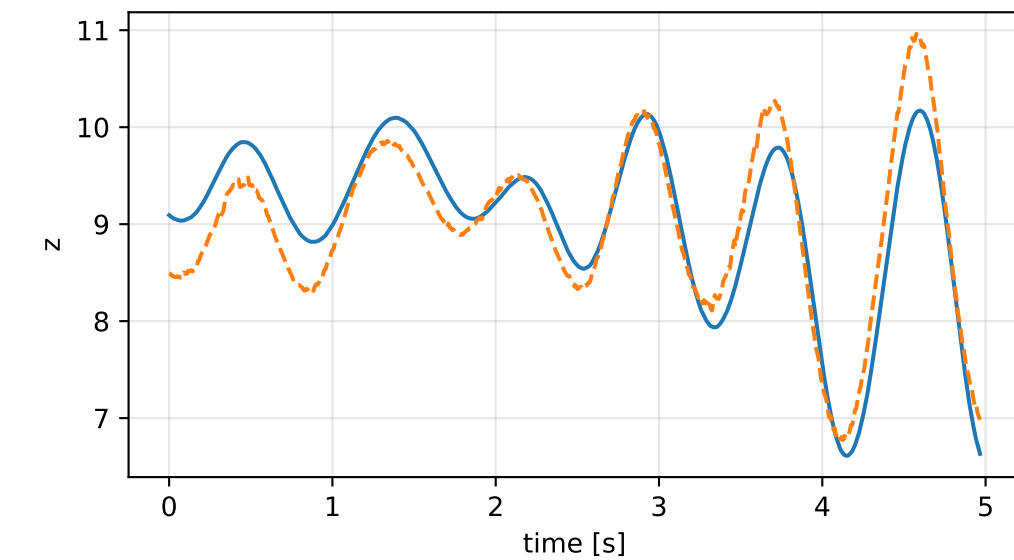
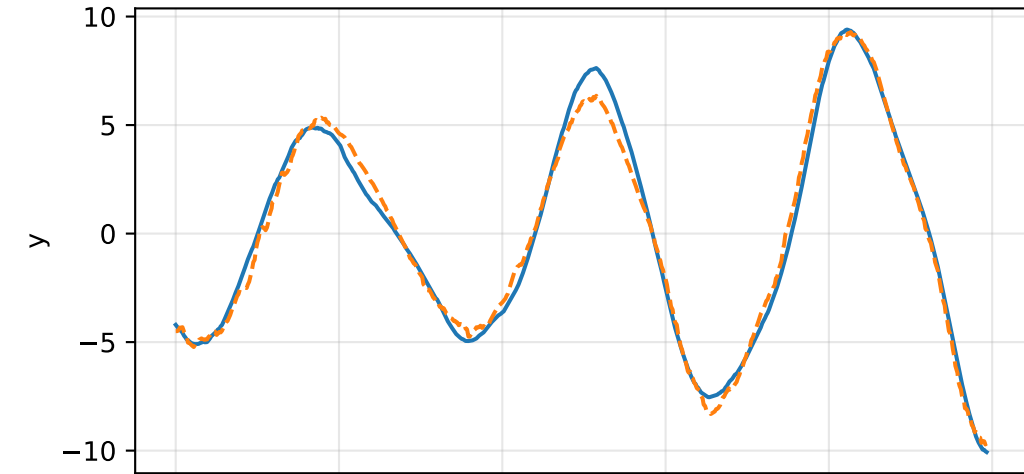
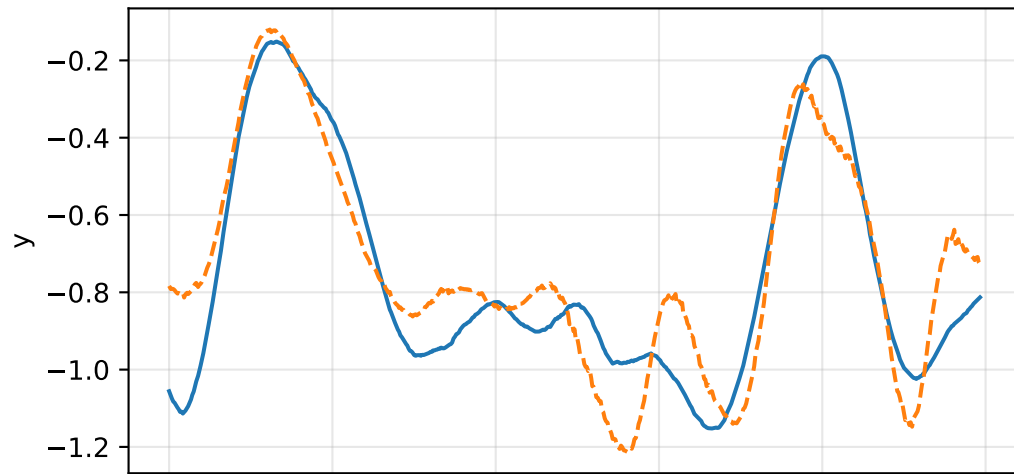
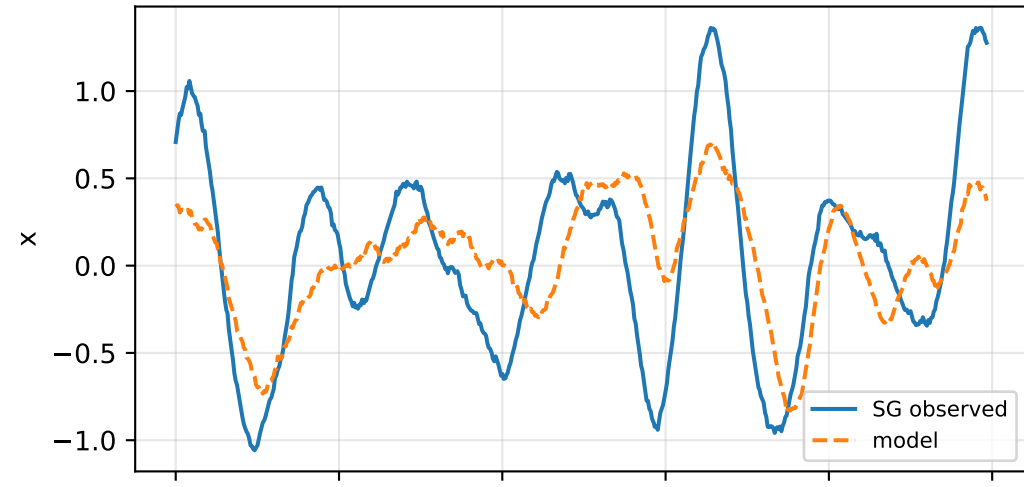


force_over_mass_rotor_3_nominal: acceleration residual objective

specific acceleration [m/s^2]

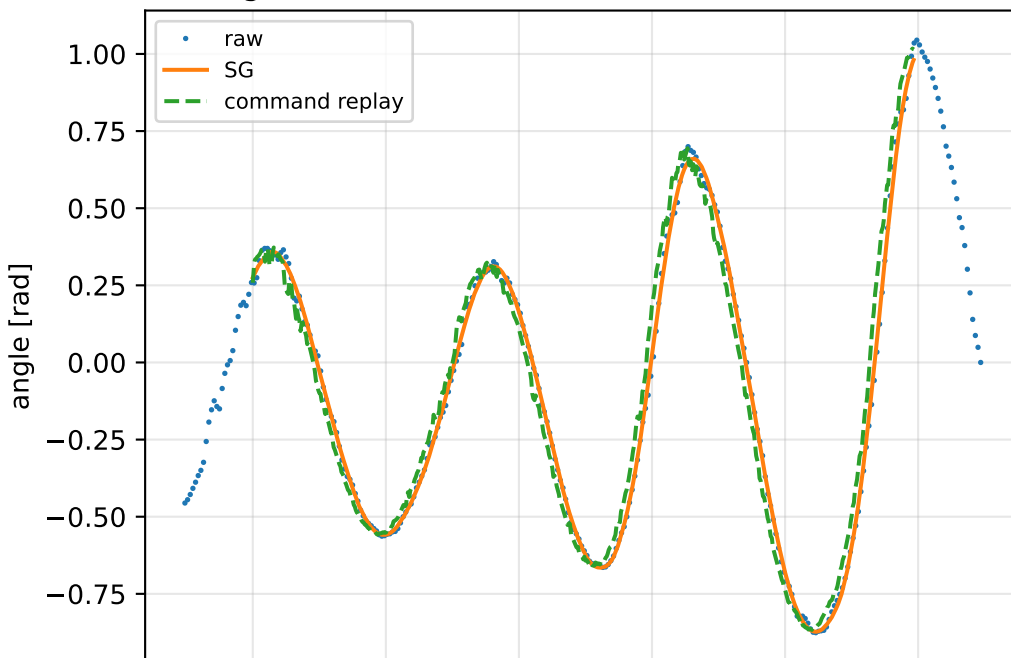


angular acceleration [rad/s^2]

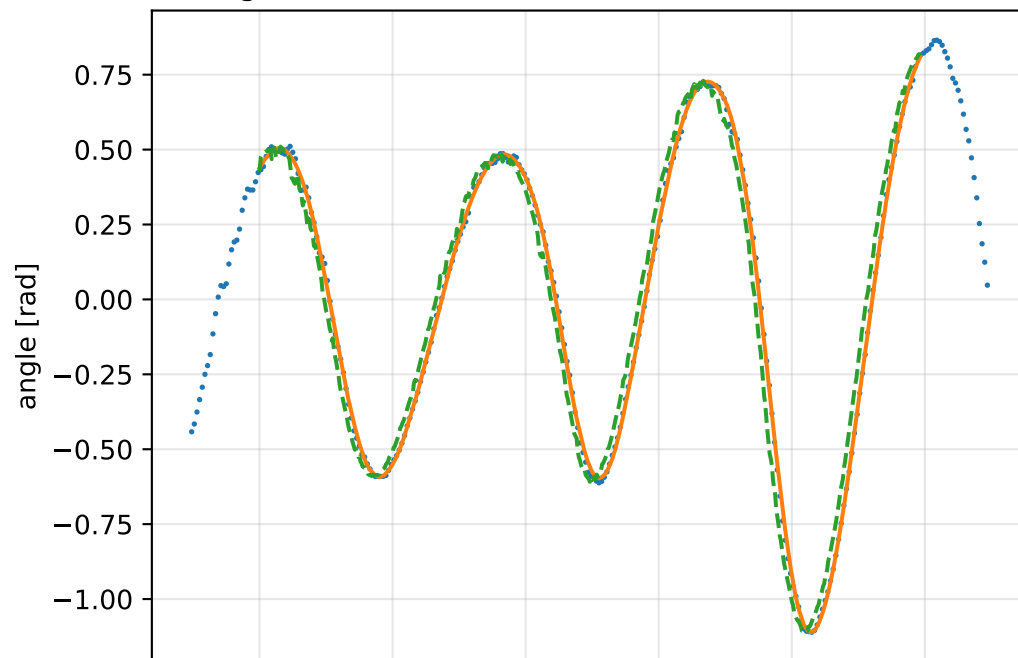


force_over_mass_rotor_3_nominal: gimbal measurement audit (objective=measured_sg)

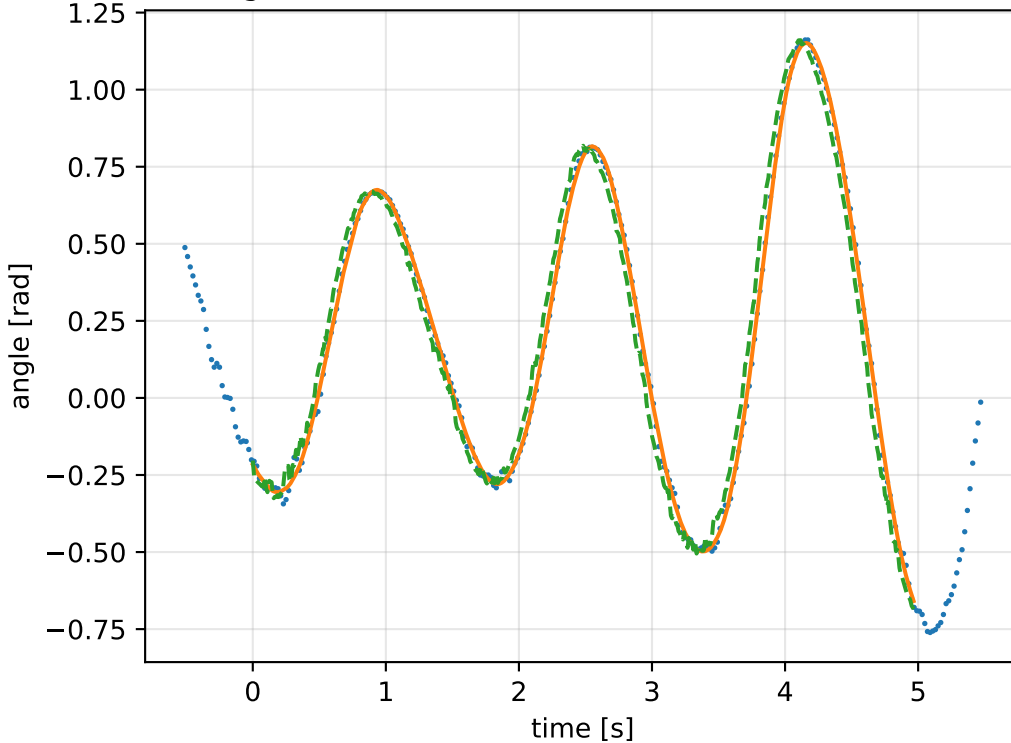
gimbal 1: RMSE=0.0759, max=0.2095 rad



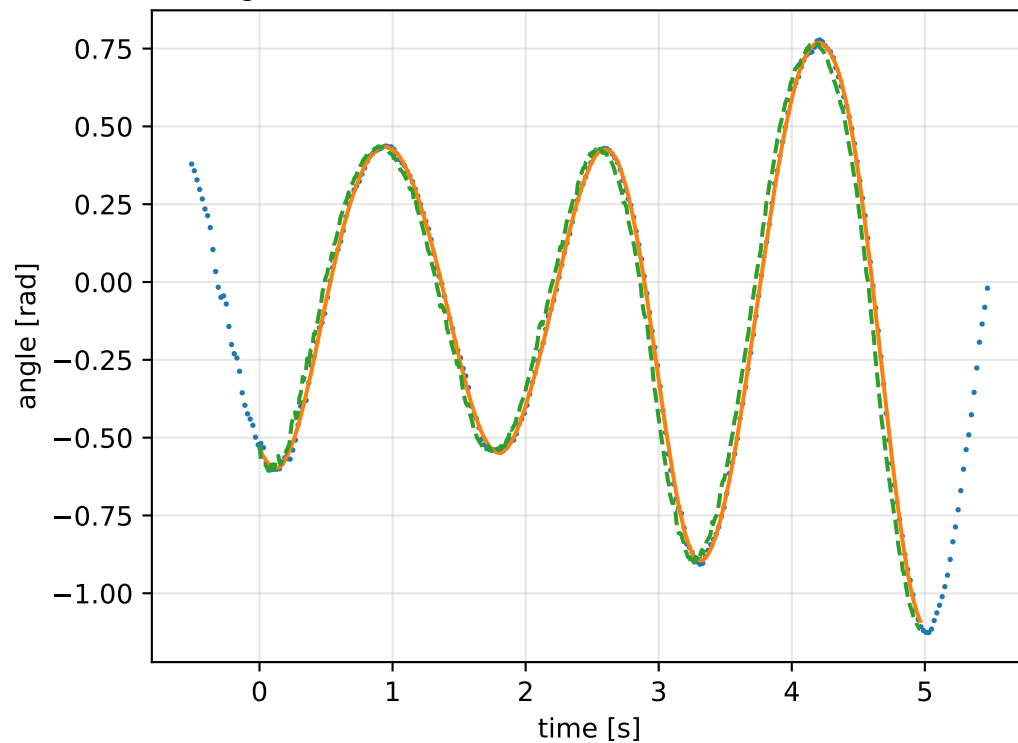
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

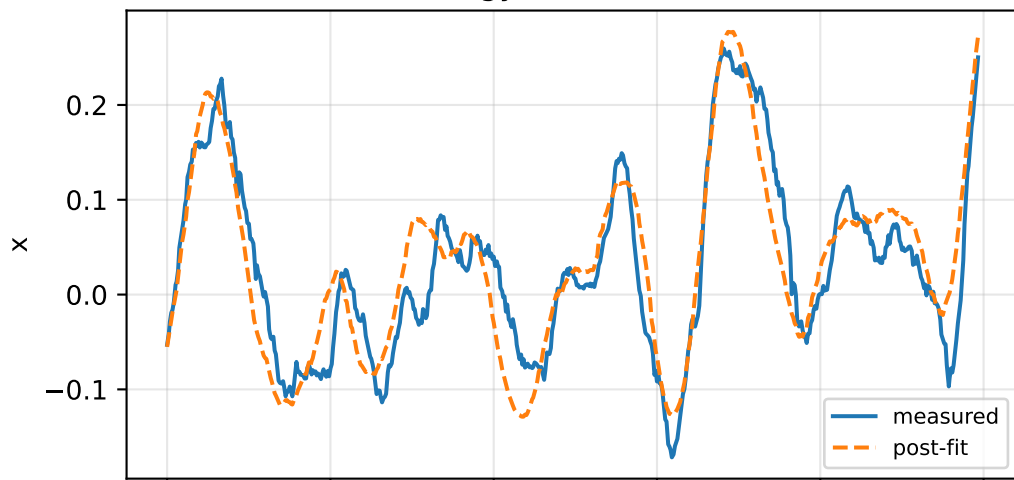


gimbal 4: RMSE=0.07131, max=0.1737 rad

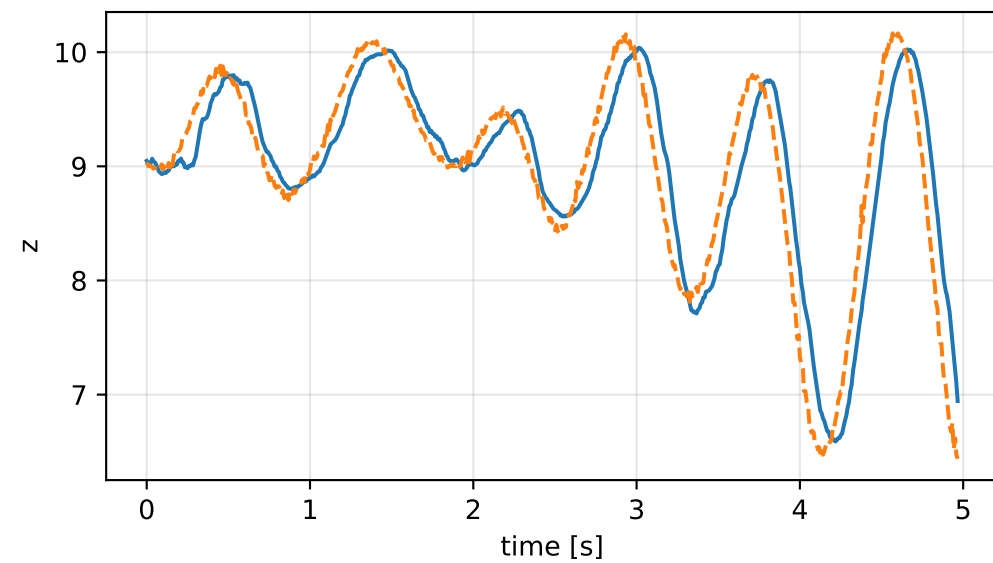
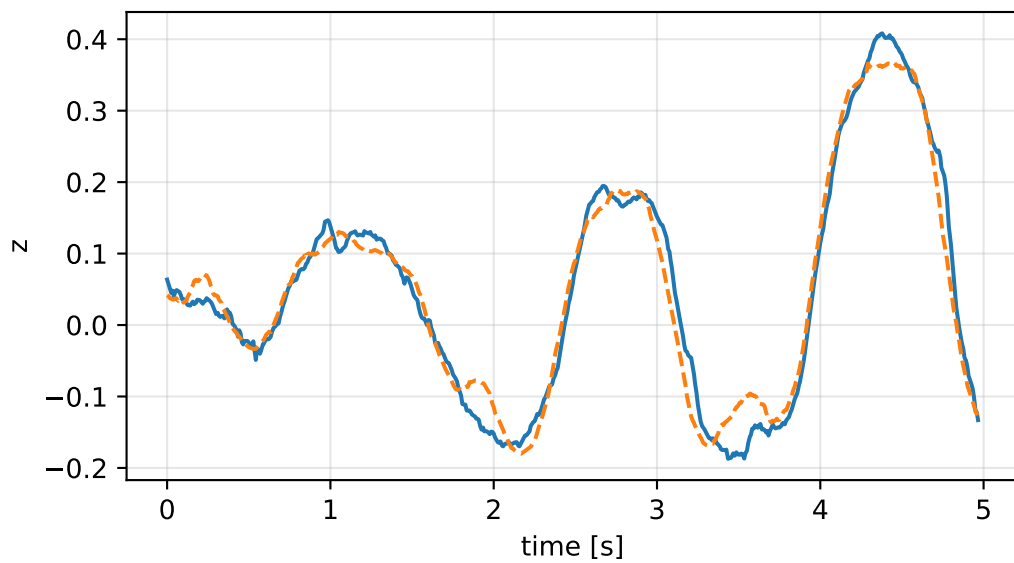
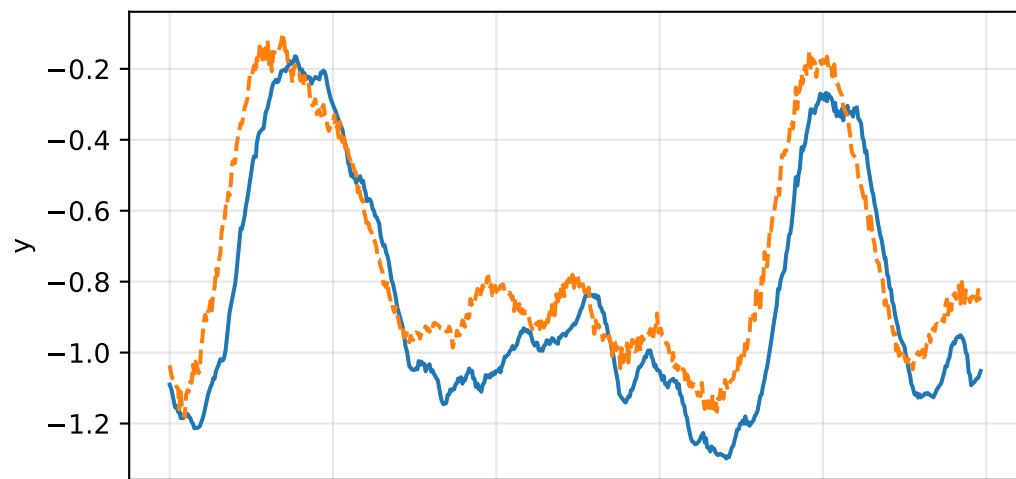
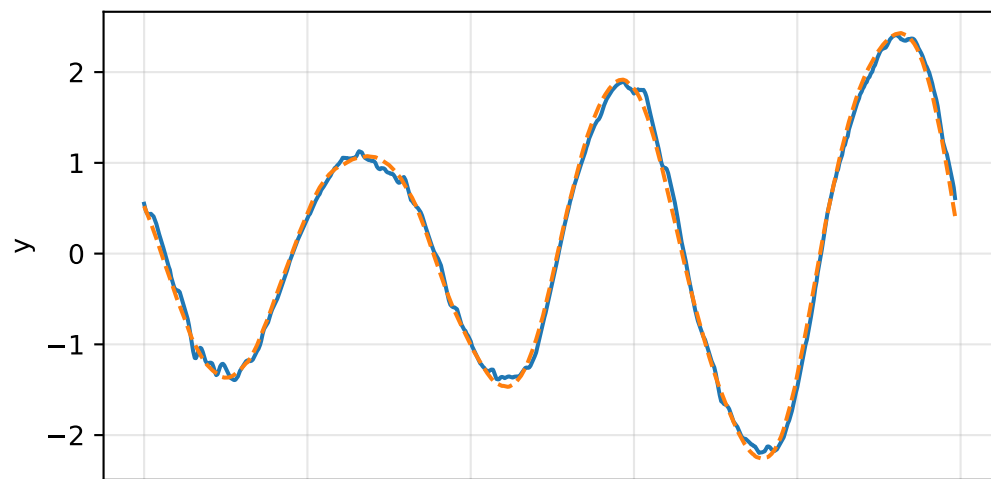
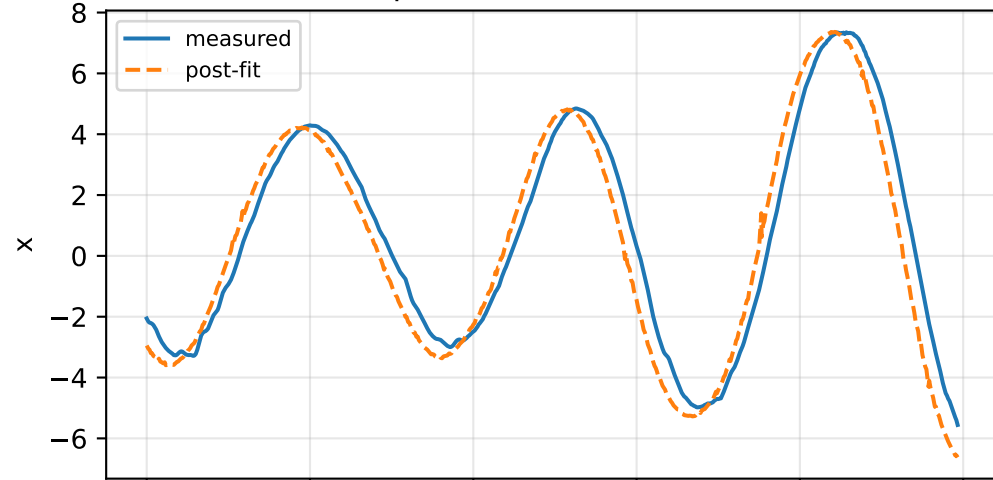


force_over_mass_rotor_3_nominal: IMU comparison (diagnostic only)

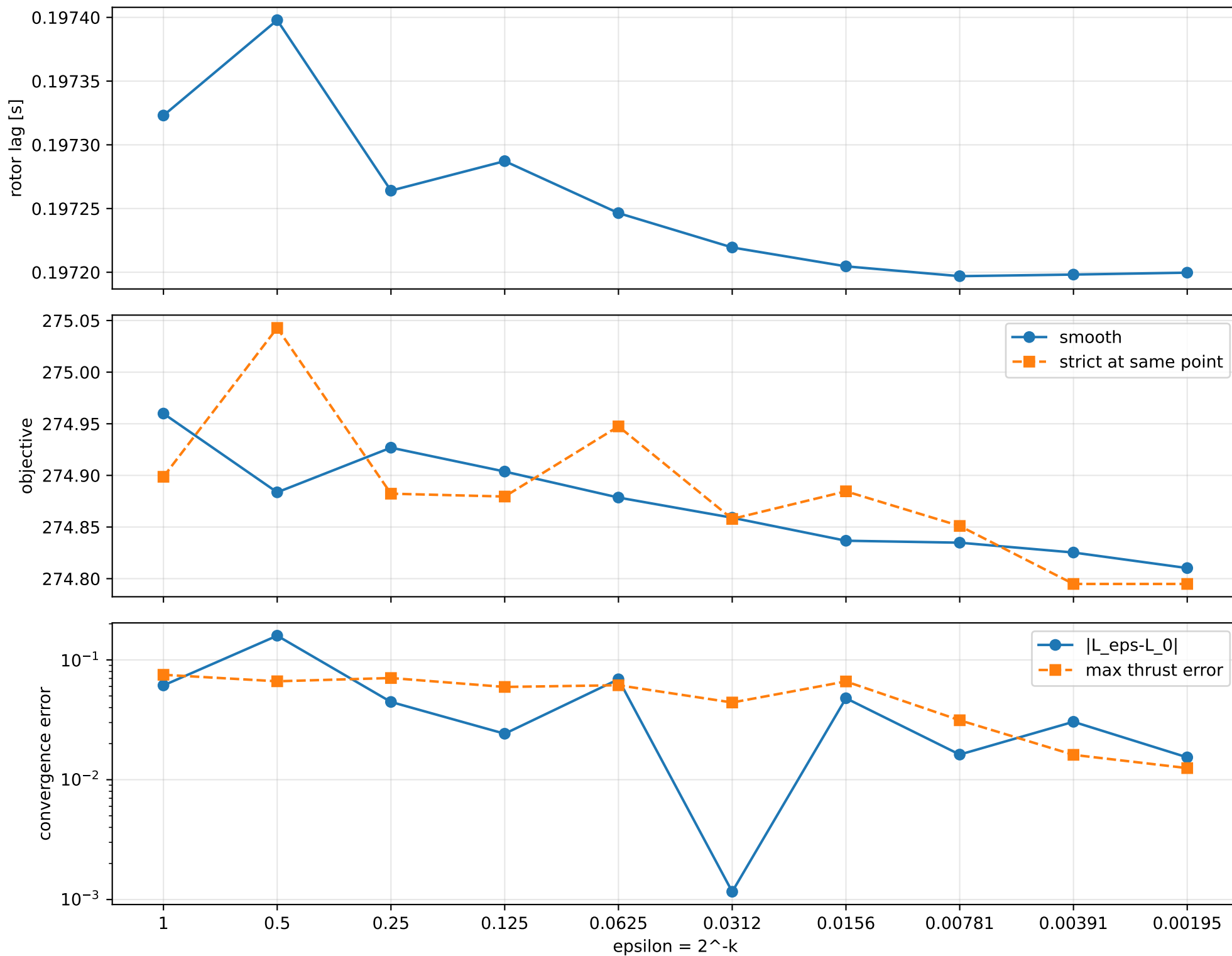
gyro [rad/s]



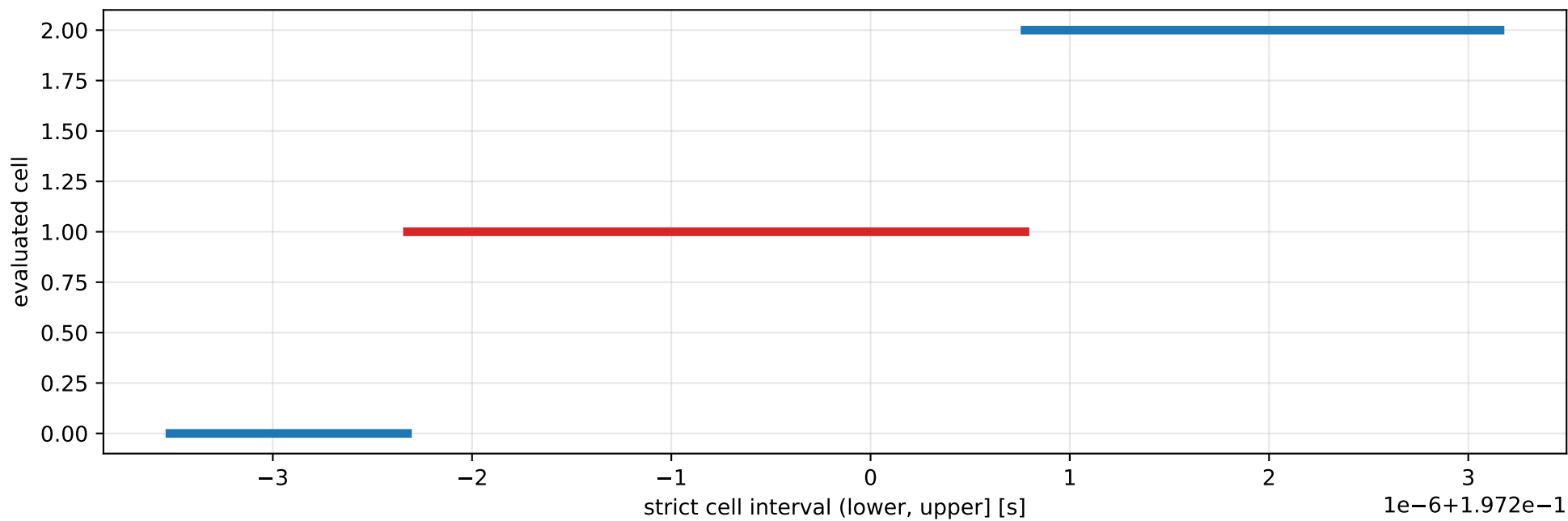
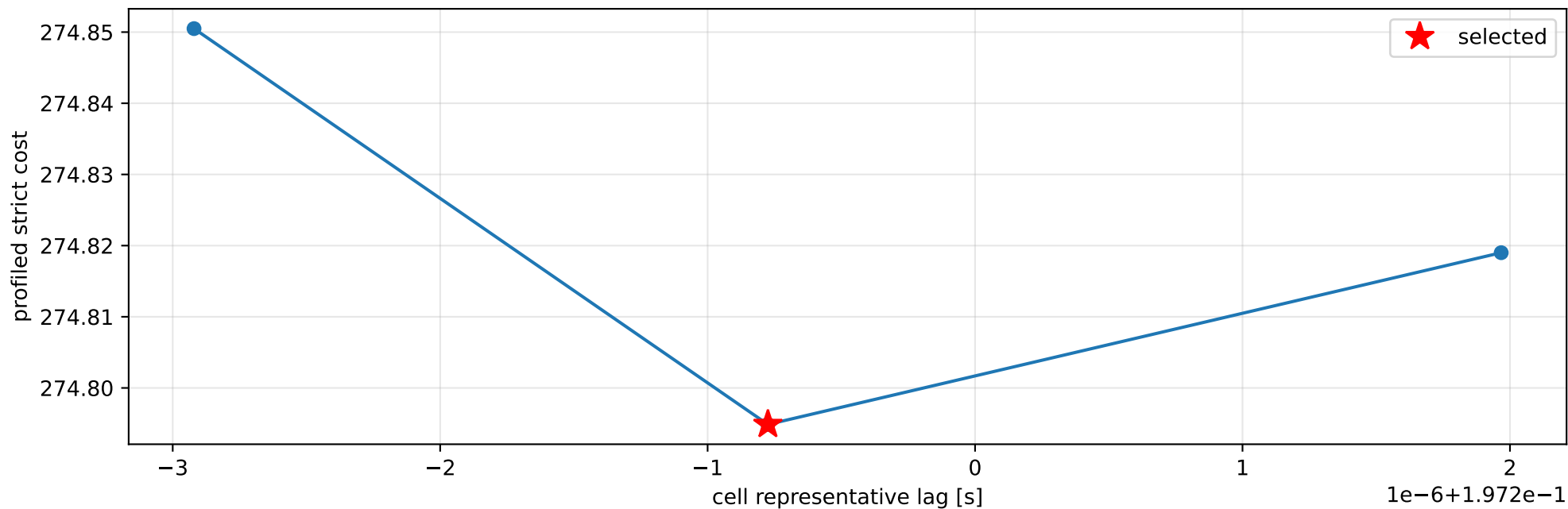
specific force [m/s²]



force_over_mass_rotor_3_nominal: smooth-to-strict continuation



force_over_mass_rotor_3_nominal: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199678 s; data boundary=False

force_over_mass_rotor_3_nominal: parameter estimate

Case: force_over_mass_rotor_3_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.76852031

inertia [kg m²]:

```
[[ 0.394241 -0.00353029 -0.04035066]
 [-0.00353029 0.28697391 -0.00854085]
 [-0.04035066 -0.00854085 0.66928743]]
```

CoG body [m]: [0.03807632 0.00087366 -0.04418264]

force effectiveness: [1.13724379 1.70082192 1.60253613 1.06043542]

scale-free J/m [m²]:

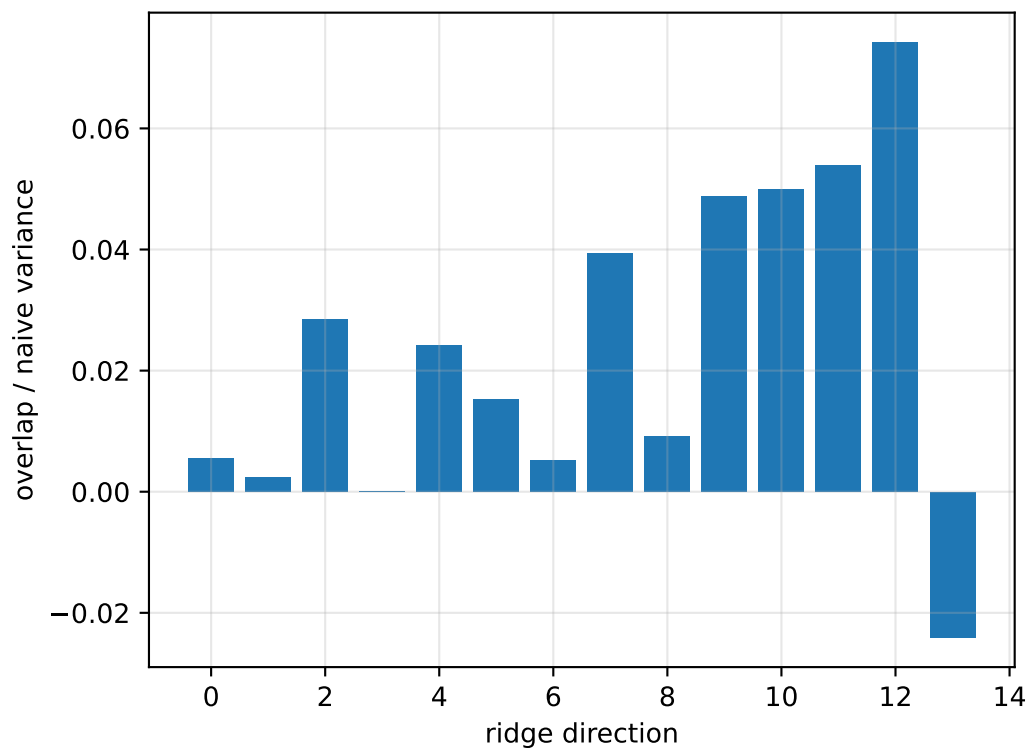
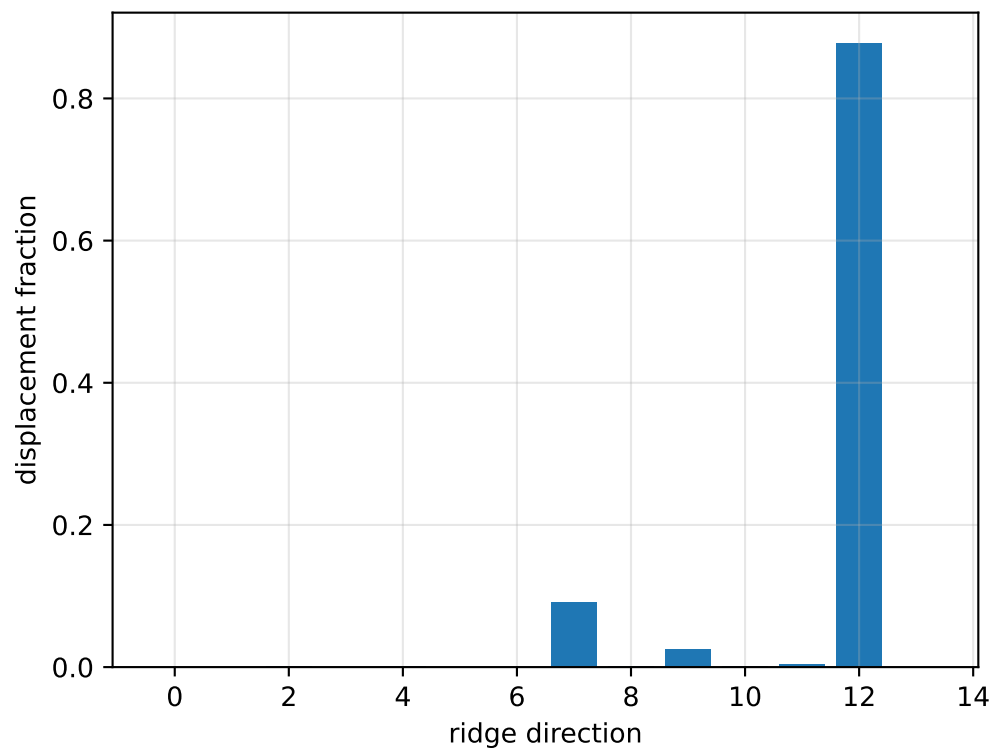
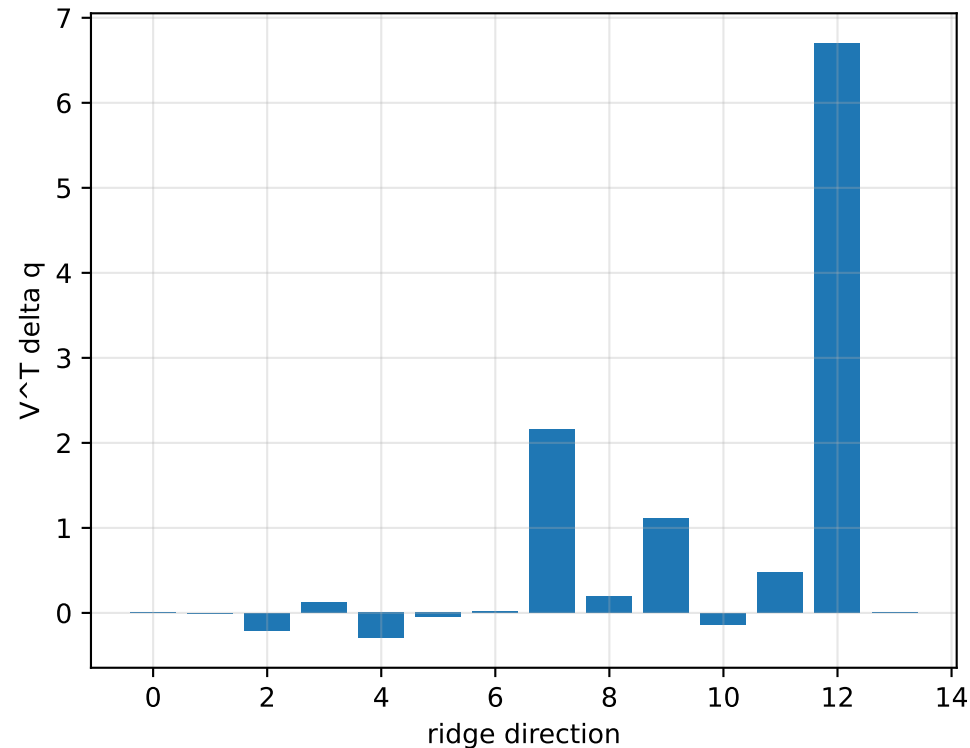
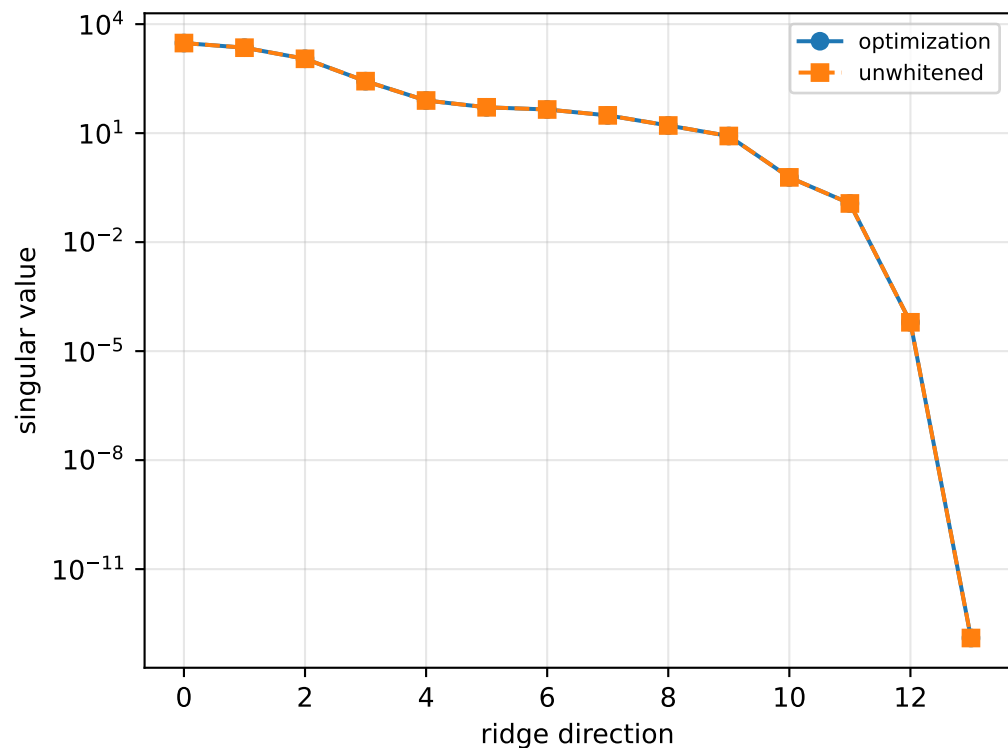
```
[[ 0.10461427 -0.00093678 -0.0107073 ]
 [-0.00093678 0.07615029 -0.00226637]
 [-0.0107073 -0.00226637 0.17759953]]
```

scale-free f/m: [0.30177462 0.45132354 0.4252428 0.28139305]

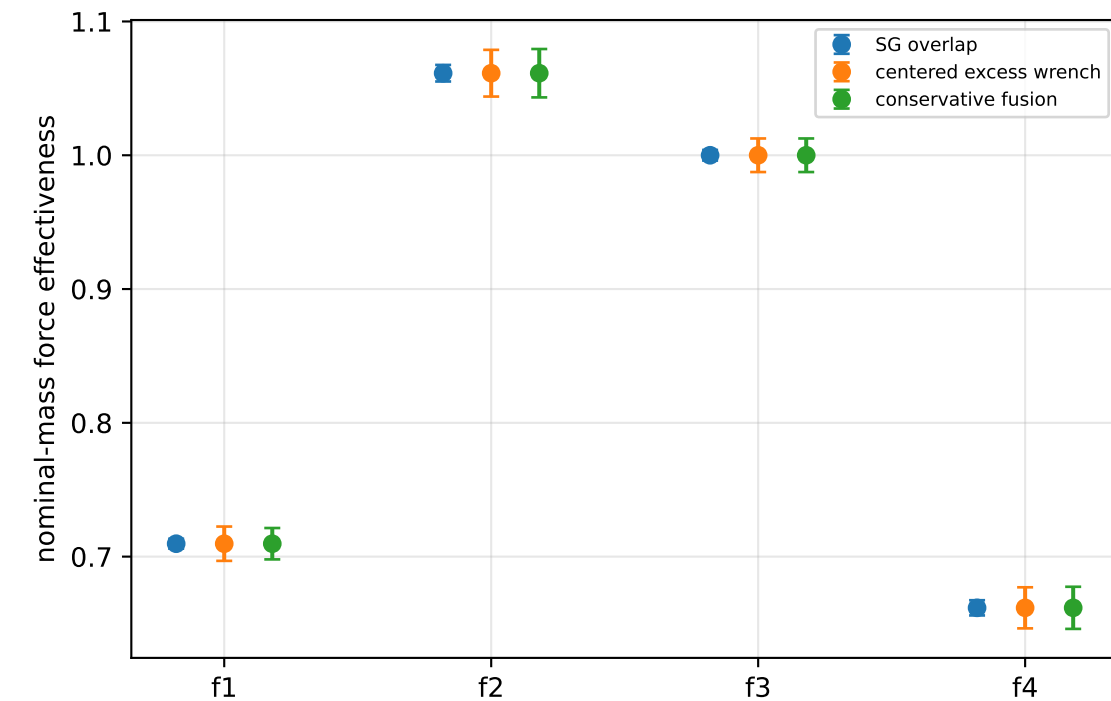
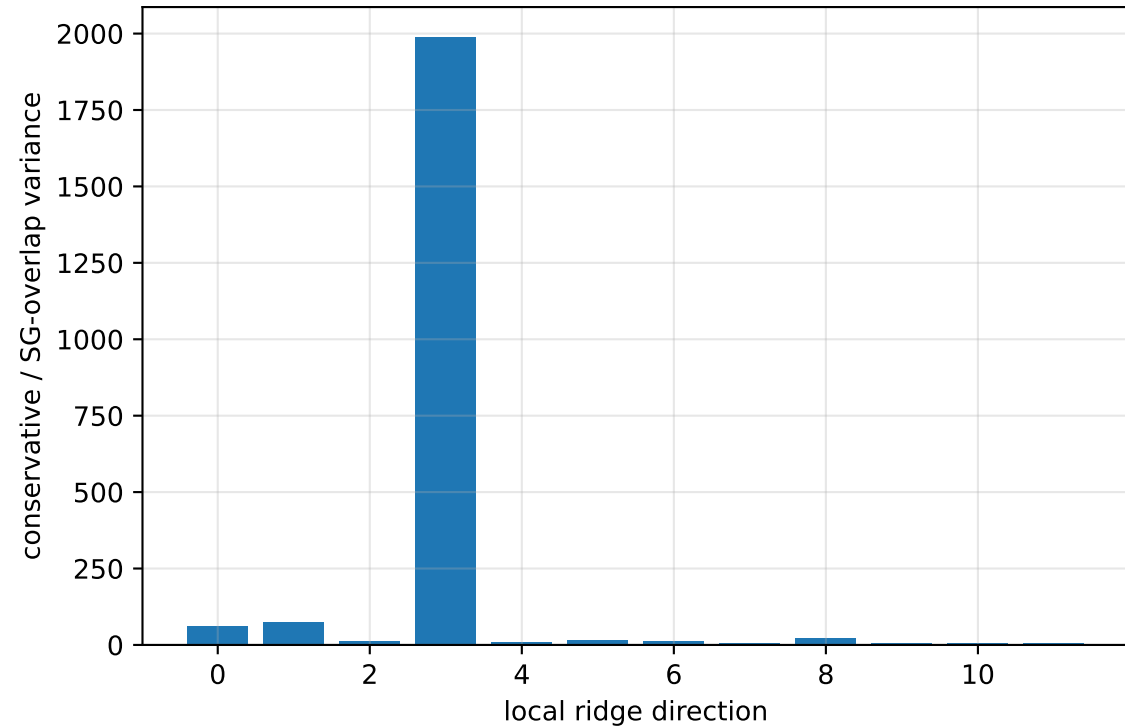
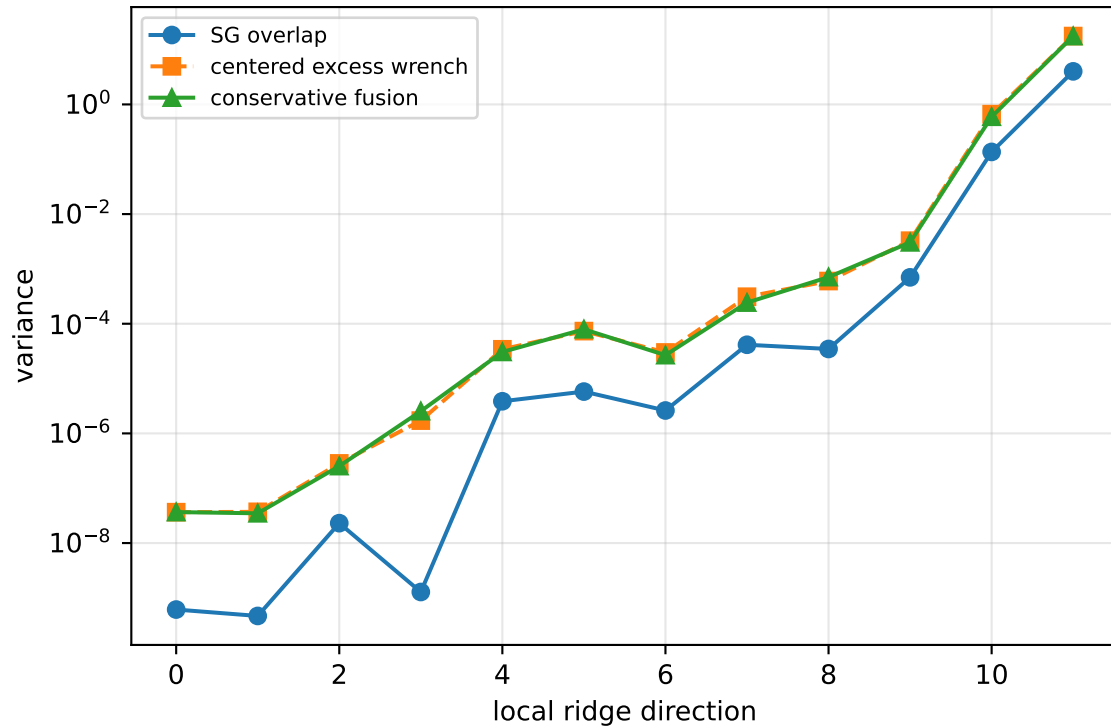
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

force_over_mass_rotor_3_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.06e-13)



force_over_mass_rotor_3_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 43.12$
wrench-to-acceleration recovery max error = $3.11\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

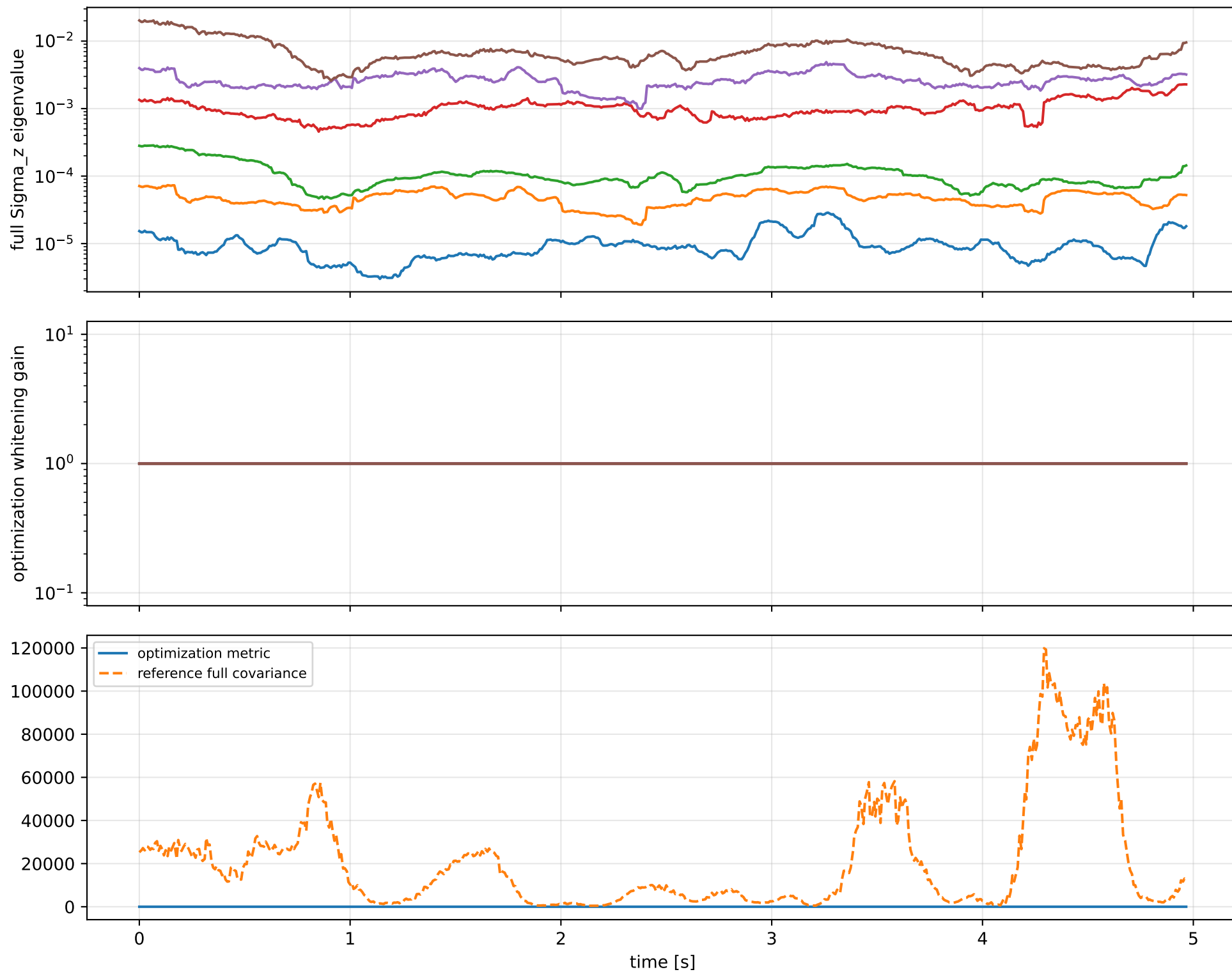
index 11: variance=17.61, ratio=4.395
 $\log_{\text{mass}} = -0.0339$; second_moment_diag_1=-0.564
second_moment_diag_2=-7.4e-05; second_moment_diag_3=+0.733
second_moment_offdiag_12=+0.0998; second_moment_offdiag_13=-0.294
second_moment_offdiag_23=-0.208; cog_x=-2.63e-05
cog_y=-0.000115; cog_z=+1.44e-05
 $\log_{\text{force_eff_1}} = -0.0334$; $\log_{\text{force_eff_2}} = -0.0335$
 $\log_{\text{force_eff_3}} = -0.0345$; $\log_{\text{force_eff_4}} = -0.0341$

index 10: variance=0.5953, ratio=4.376
 $\log_{\text{mass}} = +0.00948$; second_moment_diag_1=-0.0139
second_moment_diag_2=+0.178; second_moment_diag_3=-0.207
second_moment_offdiag_12=+0.761; second_moment_offdiag_13=+0.163
second_moment_offdiag_23=-0.565; cog_x=+0.001
cog_y=-0.00263; cog_z=+0.00035
 $\log_{\text{force_eff_1}} = +0.0163$; $\log_{\text{force_eff_2}} = +0.022$
 $\log_{\text{force_eff_3}} = +0.00435$; $\log_{\text{force_eff_4}} = -0.00891$

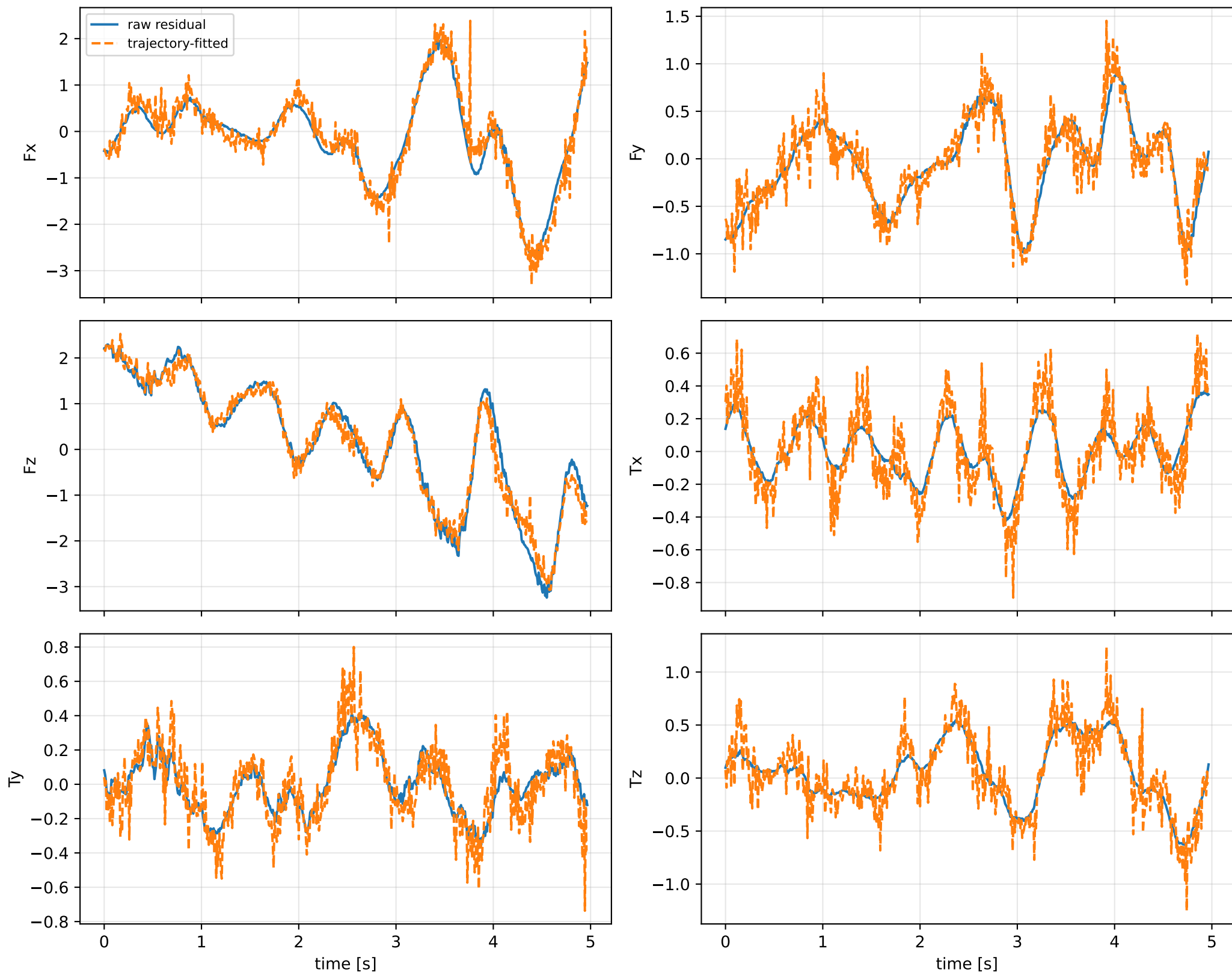
index 9: variance=0.003045, ratio=4.341
 $\log_{\text{mass}} = -0.122$; second_moment_diag_1=-0.0518
second_moment_diag_2=+0.85; second_moment_diag_3=-0.0625
second_moment_offdiag_12=-0.0767; second_moment_offdiag_13=-0.107
second_moment_offdiag_23=+0.155; cog_x=+0.036
cog_y=-0.00841; cog_z=-0.00284
 $\log_{\text{force_eff_1}} = -0.267$; $\log_{\text{force_eff_2}} = +0.032$
 $\log_{\text{force_eff_3}} = -0.00325$; $\log_{\text{force_eff_4}} = -0.375$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_3_nominal: optimization vs reference covariance



force_over_mass_rotor_3_nominal: wrench / closure (force max $1.07\text{e-}14$, torque max $8.6\text{e-}16$)



force_over_mass_rotor_3_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_3_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 6b7aab340fa6fd878752d7c8d399d9306f6b9fcfbe1938694085b3ade666560d

data objective: 274.794867289

prior objective: 3.00141272242e-06

total objective: 274.794870291

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_3_nominal (force_effectiveness_over_mass)

components: rotor_3

target: [0.42524282]

std: [1.e-05]

final: [0.4252428]

error: [-2.45006642e-08]

standardized residual: [-0.00245007]

factor objective: 3.00141272242e-06